Optimized Composition and Manufacturing Method of Natural Rubber-EPDM Composites with Nano Oxides

Abstract

Natural rubber (NR) and ethylene-propylene-diene monomer (EPDM) blends combine the excellent mechanical properties of NR with the superior aging and ozone resistance of EPDM, making them attractive for demanding applications such as conveyor belts, seals, and automotive components. However, the inherent cure incompatibility between the highly unsaturated NR and the saturated EPDM phase presents significant challenges for achieving optimal performance. This paper presents an optimized composition and manufacturing methodology for NR/EPDM composites incorporating nano-sized oxides. The optimal blend ratio of 70:30 (NR:EPDM) is established based on the balance between mechanical properties and aging resistance. Nano-zinc oxide (ZnO) serves dual functions as a curing activator and property enhancer, with 10–40 nm particles demonstrating superior efficiency compared to conventional ZnO. Nano-silica at 10 phr combined with 30 phr carbon black provides the most significant mechanical reinforcement, yielding over 117% improvement in tensile strength. Nano-alumina and nano-titania offer additional functional benefits including enhanced thermal conductivity and thermal stability. The manufacturing methodology, including mixing sequence, cure system selection, and vulcanization parameters, is described in detail. This optimized approach enables the production of high-performance NR/EPDM nanocomposites suitable for technical rubber products requiring heat resistance, alkali resistance, and dynamic durability.

Keywords: Natural rubber; EPDM; Nano-zinc oxide; Nano-silica; Nano-alumina; Cure compatibility; Nanocomposites

1. Introduction

Rubber blends have been widely used in many applications due to their compromised properties . Blends of natural rubber (NR) and ethylene-propylene-diene monomer (EPDM) have gained considerable attention because, when properly blended, NR/EPDM blends can possess good physical properties with adequately high aging and ozone resistance . NR contributes excellent

tensile strength, tear resistance, and dynamic fatigue properties, while EPDM provides outstanding resistance to ozone, oxygen, weathering, and heat aging.

However, achieving satisfactory physical properties in NR/EPDM blends is difficult due to cure incompatibility between the two rubbers . NR is highly unsaturated and chemically reactive, whereas EPDM is highly saturated and relatively non-reactive. During vulcanization, NR possesses a higher crosslinking rate, and curatives have a strong tendency to diffuse toward the NR phase . The lower solubility of many curatives in the EPDM phase further reduces crosslink formation in that phase. These factors combine to produce a blend with uneven cure distribution—the NR phase is highly crosslinked while the EPDM phase remains poorly crosslinked .

The incorporation of nano-sized oxides offers a promising route to address these challenges while simultaneously enhancing multiple performance characteristics. Nano-ZnO can replace conventional ZnO as a curing activator while improving thermal stability and dynamic mechanical properties . Nano-silica provides substantial mechanical reinforcement when properly dispersed . Nano-alumina enhances thermal conductivity and mechanical performance . Nano-titania improves thermal stability of the EPDM phase .

This paper presents a systematic framework for optimizing the composition and manufacturing method of NR/EPDM composites containing nano-sized oxides. The optimal blend ratio, nano-oxide selection and loading, cure system design, and processing parameters are established based on comprehensive evaluation of mechanical properties, cure characteristics, and thermal performance.

2. Optimized Composition Design

2.1 Blend Ratio Optimization

The NR/EPDM blend ratio is the primary determinant of the balance between mechanical properties and aging resistance. A comprehensive study on carbon-black-filled NR/EPDM

blends with varying ratios (90/10, 80/20, 70/30, 60/40, and 50/50) revealed that both tensile strength and elongation at break decrease with increasing EPDM content . However, the aging resistance improves substantially with higher EPDM fractions.

Optimal blend ratio: 70:30 (NR:EPDM)

This ratio represents the optimal compromise. At this composition, it is possible to maintain the good mechanical properties of natural rubber while improving the aging resistance of the rubber material . The 70/30 NR/EPDM ratio has been widely adopted in commercial applications and forms the basis for the compositions described in this paper.

2.2 Nano-Oxide Selection and Loading

2.2.1 Nano-Zinc Oxide (ZnO)

ZnO serves as both a curing activator and a property-enhancing filler in NR/EPDM systems. Nano-ZnO with polyhedral and spherical shapes, having dimensions ranging from 10 to 40 nm, has been synthesized and confirmed to be free of impurities .

The key advantage of nano-ZnO is its efficiency: a low content of ZnO nanoparticles is sufficient to give higher mechanical properties compared to a higher content of conventional ZnO . This is attributed to the higher surface area of nanoparticles, which provides more effective activation of the vulcanization process.

Optimal nano-ZnO content: 3–5 phr (replacing conventional ZnO at equivalent or lower loading)

2.2.2 Nano-Silica (SiO_2)

Nano-silica provides the most significant mechanical reinforcement among the nano-oxides evaluated. A systematic study on NR/EPDM (60/40) blends reinforced with nanosilica in combination with carbon black determined the appropriate contents for optimal reinforcement .

Optimal nano-silica content: 10 phr

When 10 phr nano-silica is combined with 30 phr carbon black, the resulting nanocomposite exhibits the best mechanical properties, with tensile strength enhancement exceeding 117% compared to the unreinforced blend and 40% compared to the blend reinforced with only 10 phr nano-silica .

The nanosilica used in this optimization was derived from rice husk, providing a sustainable and cost-effective source of the filler .

2.2.3 Nano-Alumina (Al_2O_3)

Nano-alumina enhances both mechanical properties and thermal conductivity of rubber composites. Research on EPDM/Al_2O_3 composites demonstrated that as Al_2O_3 content increases, mechanical properties become more excellent and thermal conductivity increases significantly .

At 100 phr loading, the thermal conductivity of the composite reaches 0.53 W/(m·K), which is approximately 32.3% higher than carbon black N550-filled composites and 57.7% higher than PVC-filled composites . The Al_2O_3 filler disperses uniformly in the rubber matrix, creating a well-developed crosslinked network that contributes to extended service life .

Recommended nano-alumina content: 5–15 phr for NR/EPDM systems (for thermal conductivity enhancement without excessive reinforcement)

2.2.4 Nano-Titania (TiO_2)

Nano-titania offers unique benefits for thermal stability enhancement. Investigation of NR/EPDM blends filled with small amounts of titania nanoparticles revealed that titania particles concentrate in the NR phase, and their presence in EPDM initiates crystal formation in the rubber phase .

Thermogravimetric analysis demonstrated improved thermal stability of EPDM and of the EPDM phase in the blend for samples containing titania nanoparticles . However, it should be noted that the presence of titania particles inhibited the vulcanization process in both rubbers, requiring adjustment of the cure system .

Recommended nano-titania content: 2–5 phr (for thermal stability enhancement with cure system adjustment)

2.3 Cure System Optimization

The cure system must be designed to address the cure incompatibility between NR and EPDM phases while accommodating the influence of nano-oxides.

2.3.1 Accelerator Selection

The accelerator type has great influence on both processability and mechanical properties of NR/EPDM blends . Comparative studies of sulfenamide, thiuram, and mercapto accelerators revealed distinct performance profiles:

TBBS (N-tert-butyl-2-benzothiazolesulfenamide) provides the best processing safety together with a relatively high state of cure. It also provides good cure compatibility between the NR and EPDM phases, giving rise to superior mechanical properties .

MBTS (2,2'-dithiobis(benzothiazole)) also gives good cure compatibility, and the tensile strength of MBTS-cured blends is relatively high. MBTS has been successfully employed as a conventional accelerator in NR/EPDM blends .

TMTD (tetramethyl thiuram disulfide) gives a high state of cure but causes severe cure incompatibility, leading to poor tensile strength .

Optimal accelerator selection: TBBS for best balance of processing safety and cure compatibility, or MBTS for good mechanical properties and cure compatibility.

2.3.2 Compatibilization Strategies

Several strategies can further improve cure compatibility and mechanical performance:

Mercapto-modified EPDM (EPDMSH): The mercapto groups react with double bonds of the NR phase, decreasing interfacial energy. As low as 2.5 phr of this functionalized copolymer achieves better mechanical performance. EPDMSH acts as an accelerator, decreasing both scorch time and optimum curing time .

Maleic anhydride compatibilizer: Maleic anhydride can be used as a compatibilizer in EPDM/NR blends, with binary accelerator systems (MBTS/TMTD) providing the best aging resistance and compatibility .

Thioacetate-modified EPDM (EPDMTA): Also acts as an accelerator agent, indicated by higher cure rate index and lower activation energy of vulcanization .

2.4 Complete Optimized Formulation

Based on the comprehensive analysis, the following optimized formulation is proposed for NR/EPDM nanocomposites:

Component	Content (phr)	Function
Natural Rubber (SMR 5L or STR 5L)	70	Matrix, mechanical strength
EPDM (ENB type, 4–5% ENB)	30	Aging and ozone resistance
Nano-ZnO (10–40 nm)	4	Curing activator
Stearic acid	2	Co-activator
Nano-silica (modified)	10	Mechanical reinforcement
Carbon black (N330 or N550)	30	Primary reinforcement
Nano-alumina (optional)	5–10	Thermal conductivity
Process oil	5	Processing aid
TBBS	1.2	Primary accelerator
MBTS	0.5	Secondary accelerator
Sulfur	1.5–2.0	Crosslinking agent
Antioxidant (TMQ)	1.0	Aging protection

3. Manufacturing Methodology

3.1 Mixing Process

The mixing sequence significantly influences the dispersion of nano-oxides and the final properties of the blend. Two primary approaches have been established: conventional single-stage mixing and sequential phase mixing.

3.1.1 Conventional Two-Roll Mill Mixing

The traditional mixing procedure involves the following steps :

Step 1: Mastication of NR

Natural rubber is passed through a laboratory two-roll mill with starting roll temperature of approximately 80°F (27°C). The nip gap is adjusted to form a band on the front roll.

Step 2: Incorporation of EPDM

High molecular weight EPDM is blended with the NR. After the NR begins to soften and roll temperatures start to rise, the EPDM is added. Blending continues until a uniform NR/EPDM blend is achieved .

Step 3: Addition of Ingredients

Nano-oxides (ZnO, silica, etc.), stearic acid, and fillers are added to the blend. The process oil is added at the end of the blending process .

Step 4: Cooling and Curative Addition

The compounded masterbatch is allowed to cool to ambient temperature, after which curatives and accelerators are added .

3.1.2 Sequential Phase Mixing

For improved dispersion of nano-oxides and better cure compatibility, sequential phase mixing is recommended :

Phase 1: EPDM is mixed with the particulate reinforcing filler (carbon black, nano-silica, nano-oxides) to create a first rubber composition containing well-dispersed fillers.

Phase 2: The first composition is blended with natural rubber (cis 1,4-polyisoprene) to form the second rubber composition.

Phase 3: Sulfur curative is blended with the second rubber composition.

This approach results in better carbon black reinforcement of the EPDM portion and improved overall resistance to dynamic flex failure .

3.1.3 Melt Compounding in Internal Mixer

For larger-scale production, melt blending in a Brabender internal mixer or equivalent equipment is preferred. The appropriate contents of nano-silica (10 phr) and carbon black (30 phr) for reinforcing NR/EPDM nanocomposites were established using this method .

3.2 Vulcanization Process

3.2.1 Cure Characteristics

The cure characteristics of the optimized compound should be determined using an oscillating disk rheometer (ODR) or moving die rheometer (MDR) at the intended vulcanization temperature (typically 150–160°C for NR/EPDM systems).

The addition of nano-ZnO results in a better state of curing and higher maximum torque with a significant decrease in optimum curing time and scorching time . This indicates that nano-ZnO accelerates the vulcanization process, requiring adjustment of cure times accordingly.

3.2.2 Vulcanization Conditions

For the optimized NR/EPDM nanocomposite formulation:

Temperature: 150–160°C (for compression molding)

Pressure: 10–15 MPa

Time: Determined by rheometer tc90 (typically 15–30 minutes depending on thickness)

For autoclave vulcanization, conditions of 300°F (149°C) and 32 psig for one hour have been successfully employed for NR/EPDM blends .

3.3 Post-Vulcanization and Testing

After vulcanization, samples should be aged at room temperature for approximately 24 hours before testing to allow for equilibration of properties .

4. Performance Characteristics

4.1 Mechanical Properties

The optimized NR/EPDM nanocomposite with 10 phr nano-silica and 30 phr carbon black exhibits tensile strength enhancement exceeding 117% compared to unreinforced blends and 40% improvement compared to blends with only nano-silica reinforcement . The modulus and hardness increase with higher EPDM content, while tensile strength and elongation at break decrease .

4.2 Thermal Properties

Nano-ZnO improves the thermal stability and dynamic mechanical properties of NR/EPDM blends . Nano-alumina significantly enhances thermal conductivity, reaching 0.53 W/(m·K) at 100 phr loading . Nano-titania improves thermal stability of the EPDM phase .

4.3 Aging and Ozone Resistance

The 70/30 NR/EPDM blend ratio provides substantial improvement in aging resistance compared to pure NR. The addition of nano-ZnO further enhances thermo-mechanical performance . Binary accelerator systems (MBTS/TMTD) provide the best aging resistance and blend compatibility .

4.4 Swelling Resistance

Modified nanosilica has been shown to improve swelling resistance of NR/EPDM blends in addition to enhancing curing and mechanical properties .

5. Conclusions

This paper has presented an optimized composition and manufacturing methodology for NR/EPDM composites incorporating nano-sized oxides. The key conclusions are:

Composition optimization:

- The optimal NR:EPDM blend ratio is 70:30, balancing mechanical properties with aging resistance.

- Nano-ZnO (10–40 nm, 4 phr) serves as an efficient curing activator, enabling reduction in ZnO loading.

- Nano-silica (10 phr) combined with carbon black (30 phr) provides the most significant mechanical reinforcement (117% tensile strength improvement).

- Nano-alumina (5–10 phr) enhances thermal conductivity.

- Nano-titania (2–5 phr) improves thermal stability of the EPDM phase.

Manufacturing methodology:

- Sequential phase mixing, where EPDM is first compounded with fillers before blending with NR, improves filler dispersion and cure compatibility.

- TBBS or MBTS accelerators provide the best balance of processing safety and cure compatibility.

- Mercapto-modified EPDM (2.5 phr) serves as an effective reactive compatibilizer.

- Vulcanization at 150–160°C with cure times determined by rheometer tc90 yields optimal properties.

Performance outcomes:

- Tensile strength enhancement exceeding 117% compared to unreinforced blends.

- Thermal conductivity improvement with nano-alumina addition.

- Enhanced thermal stability with nano-ZnO and nano-titania.

- Improved aging resistance and swelling resistance.

The optimized NR/EPDM nanocomposite formulation is suitable for technical rubber products requiring heat resistance, alkali resistance, abrasion resistance, and low endogenous heat generation, such as conveyor belts for cement industry applications and other demanding environments .

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